Identification and Control of a Crane System

Eric Castro Velásquez*, Esteban Segura Garcia*, Daniel Chavarria Garcia*
*School of Electronics Engineering, Instituto Tecnológico de Costa Rica (ITCR),
30101 Cartago, Costa Rica
{ecastro, essegura, dachavarria}@estudiantec.cr

***Abstract*—This paper presents the experimental identification and modern state-feedback control of such a system. Empirical transfer functions for the cart position and pendulum angle were identified from physical plant data using the MATLAB System Identification Toolbox and combined into a fourth-order SIMO state-space representation. Two Integral State Feedback controllers were designed and evaluated: one based on Pole Placement and one on the Linear Quadratic Regulator (LQR) criterion, with performance assessed in terms of settling time, overshoot, steady-state error, and pendulum oscillation attenuation.Both controllers achieved zero steady-state error under a $0.5\,\mathrm{m}$ step reference. However, LQR significantly outperformed Pole Placement in transient response and structural safety: while Pole Placement produced a pendulum deflection of $0.20\,\mathrm{rad}$, exceeding the $\pm 0.175\,\mathrm{rad}$ safety bound, LQR constrained the deflection to $0.17\,\mathrm{rad}$ and eliminated cart overshoot entirely ($M_p = 0\%$) with a settling time penalty of only 1.7%. These results establish LQR with integral action as the preferred framework for balancing tracking performance and physical safety constraints in underactuated crane systems.**

I. Introduction

II. System Identification

A. Experimental Setup

The experimental platform consists of a cart driven by a DC motor through a pulley-belt transmission system. A pendulum is attached to the cart and both cart position and pendulum angle are measured using incremental optical encoders.

The plant was excited using a trapezoidal voltage input of approximately $7.5\,\mathrm{V}$. Experimental responses were sampled every $20\,\mathrm{ms}$.

B. Position Transfer Functions

The identified cart-position transfer functions obtained from the selected experiments were:

$$G_{p1}(s) = \frac{0.2493}{s^2 + 10.41s + 7.838 \times 10^{-7}} \tag{1}$$

$$G_{p2}(s) = \frac{0.2542}{s^2 + 10.5s + 1.146 \times 10^{-6}} \tag{2}$$

$$G_{p5}(s) = \frac{0.2546}{s^2 + 10.44s + 8.605 \times 10^{-7}} \tag{3}$$

$$G_{p6}(s) = \frac{0.2611}{s^2 + 10.69s + 6.719 \times 10^{-7}} \tag{4}$$

Table I summarizes the extracted parameters.

TABLE I
Position Model Parameters

Model	K	a_x
grua1	0.2493	10.41
grua2	0.2542	10.50
grua5	0.2546	10.44
grua6	0.2611	10.69

C. Angular Transfer Functions

The identified pendulum-angle transfer functions were:

$$G_{\theta 1}(s) = \frac{-0.3494s - 0.4978}{s^2 + 0.01909s + 35.34} \tag{5}$$

$$G_{\theta 2}(s) = \frac{-0.3951s - 0.3945}{s^2 + 0.01639s + 35.33} \tag{6}$$

$$G_{\theta 5}(s) = \frac{-0.4017s - 0.4771}{s^2 + 0.01663s + 35.36} \tag{7}$$

$$G_{\theta 6}(s) = \frac{-0.3969s - 0.5118}{s^2 + 0.01526s + 35.36} \tag{8}$$

The corresponding parameters are summarized in Table II.

TABLE II
Angular Model Parameters

Model	ω_n	ζ	K
grua1	5.9447	0.001606	-0.0141
grua2	5.9439	0.001379	-0.0112
grua5	5.9464	0.001398	-0.0135
grua6	5.9464	0.001283	-0.0145

D. Average SIMO Model

Average transfer functions were obtained from the identified models:

$$G_p(s) = \frac{0.2548}{s^2 + 10.51s + 8.66 \times 10^{-7}} \tag{9}$$

$$G_\theta(s) = \frac{-0.3858s - 0.4703}{s^2 + 0.01685s + 35.35} \tag{10}$$

Using the state vector:

$$x^T = \begin{bmatrix} p & \theta & \dot{p} & \dot{\theta} \end{bmatrix} \tag{11}$$

the resulting SIMO state-space model becomes:

$$\dot{x} = Ax + Bu \tag{12}$$

with:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -10.51 & 0 \\ 0 & -35.35 & 0 & -0.0169 \end{bmatrix} \tag{13}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0.2548 \\ -0.3858 \end{bmatrix} \tag{14}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \tag{15}$$

III. Controller Design

An Integral State Feedback strategy was implemented to eliminate steady-state error while regulating the cart position.

The augmented system was constructed by adding an integral state associated with the position tracking error.

A. Pole Placement Design

To guarantee that the closed-loop system dynamics surpass the minimum nominal requirements ($t_s < 5$ s and $M_p < 5\%$), a more aggressive design target was selected with a design settling time $T_s = 3.7$ s and a target overshoot $M_p = 0.03$ (3%). Based on these specifications, the damping ratio ζ and natural frequency ω_n were computed to establish the ideal location for the dominant closed-loop poles.

The augmented system requires the allocation of five poles due to the integration of the position tracking error state (x_I). To ensure that the transient behavior is strictly governed by the intended second-order response, the three remaining non-dominant poles were positioned along the real axis at a safe distance at least four times further to the left than the real part of the dominant poles ($\sigma = -\zeta\omega_n$). This configuration prevents the non-dominant dynamics from introducing significant slow tails or undesirable couplings in the time response. The selected closed-loop pole are defined as:

$$\begin{aligned} p_{1,2} &= -1.0811 \pm 0.9686j \quad \text{(Dominant Pairs)} \\ p_3 &= -4.3243 + 0.0000j \quad \text{(Non-Dominant, } 4\sigma) \\ p_4 &= -5.4054 + 0.0000j \quad \text{(Non-Dominant, } 5\sigma) \\ p_5 &= -6.4865 + 0.0000j \quad \text{(Non-Dominant, } 6\sigma) \end{aligned}$$

By applying Ackermann's formula via the `place` command in MATLAB, the optimal state-feedback gain vector K and the integral tracker gain K_i were obtained as follows:

$$K = \begin{bmatrix} 56.6091 & -17.8418 & 2.2805 & -0.8231 \end{bmatrix}$$
$$K_i = 35.4650$$

B. LQR Design

For the LQR controller, the system was augmented with an integral state:

$$\dot{x}_I = r - C_r x \tag{16}$$

where:

$$C_r = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \tag{17}$$

The augmented system matrices become:

$$A_{aug} = \begin{bmatrix} A & 0 \\ -C_r & 0 \end{bmatrix}, \qquad B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix} \tag{18}$$

The controllability matrix of the augmented system presented rank 5, confirming full controllability.

The LQR minimizes the quadratic cost:

$$J = \int_0^\infty \left(x^T Q x + u^T R u \right) dt \tag{19}$$

Weighting matrices were selected using Bryson's rule, setting each diagonal entry as the inverse square of the maximum allowable deviation for the corresponding state:

$$Q = \text{diag}\left(\frac{1}{p_{\max}^2}, \frac{1}{\theta_{\max}^2}, \frac{1}{\dot{p}_{\max}^2}, \frac{1}{\dot{\theta}_{\max}^2}, \frac{1}{\xi_{\max}^2} \right) \tag{20}$$

$$R = \frac{1}{u_{\max}^2} \tag{21}$$

The maximum allowable values used were:

TABLE III
Bryson's Rule Design Parameters

State	Symbol	Max. value	Q_{ii}
Position	p	$0.112\,\mathrm{m}$	80.0
Angle	θ	$0.209\,\mathrm{rad}$	23.0
Linear vel.	$\dot{p}$	$1.0\,\mathrm{m/s}$	1.0
Angular vel.	$\dot{\theta}$	$0.354\,\mathrm{rad/s}$	8.0
Integral error	ξ	$0.158\,\mathrm{m\cdot s}$	40.0
Control input	u	$1.0\,\mathrm{V}$	$R = 1.0$

The resulting augmented gain vector was:

$$\hat{K}_{LQR} = \begin{bmatrix} 80.0354 & -23.1906 & 7.1329 & -7.0107 & -40.0000 \end{bmatrix} \tag{22}$$

Thus, the state-feedback and integral gains were:

$$K = \begin{bmatrix} 80.0354 & -23.1906 & 7.1329 & -7.0107 \end{bmatrix} \tag{23}$$

$$K_I = -40.0 \tag{24}$$

The obtained closed-loop poles were:

$$\lambda_{cl} = \{-10.4662,\ -1.4987 \pm 6.1233j,\ -0.7928 \pm 0.4876j\} \tag{25}$$

IV. Results and Discussion

A. Pole Placement Controller

The Pole Placement controller was first validated in simulation by applying a $0.5\,\mathrm{m}$ step reference to the closed-loop system. As shown in Fig. 1, the cart position response satisfied all primary design constraints, yielding a settling time of $t_s = 4.4995\,\mathrm{s}$, a peak overshoot of $M_p = 2.42\%$, and zero steady-state error, confirming the effectiveness of the integral state feedback augmentation. However, the pendulum angle reached a peak deflection of $-0.26\,\mathrm{rad}$ ($14.89°$), exceeding the $\pm 0.175\,\mathrm{rad}$ ($\pm 10°$) structural safety bound. This violation arises because pole placement optimizes closed-loop dynamics based solely on eigenvalue location, without explicitly penalizing inter-axis energy coupling or bounding intermediate state trajectories.

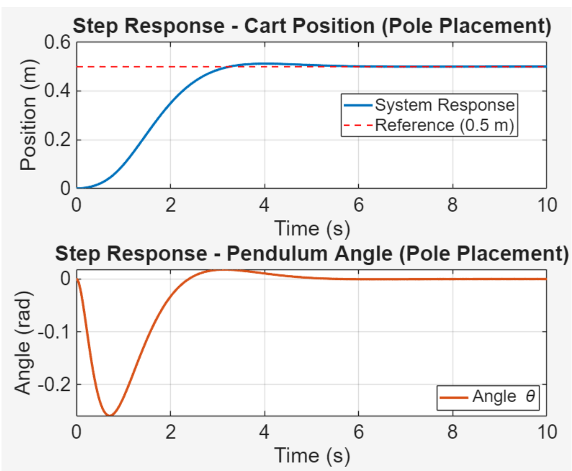

Fig. 1. Simulated closed-loop step response for cart position and pendulum angle under Pole Placement control.

The controller was subsequently deployed on the physical platform. The experimental cart position and pendulum angle profiles are shown in Fig. 2 and Fig. 3, respectively. The measured response yielded a settling time of $t_s = 5.16\,\mathrm{s}$ and an overshoot of $M_p = 3.6\%$, with zero steady-state error preserved by the integral action. The increase in both metrics relative to simulation is attributed to unmodeled nonlinearities, including joint friction, motor dead-zones, sensor noise, and the processing latency of the $20\,\mathrm{ms}$ sampling interval. These discrepancies motivate the adoption of an optimal control strategy capable of explicitly trading off tracking speed against parametric uncertainty and state constraints.

B. LQR Controller

The LQR controller was evaluated under the same $0.5\,\mathrm{m}$ step reference. In simulation, the cart position response, shown in Fig. 4, achieved a settling time of $t_s = 4.701\,\mathrm{s}$, a peak overshoot of $M_p = 0.601\%$, and zero steady-state error. The corresponding pendulum angle response, shown in Fig. 5, remained within the safety bound throughout the transient, with a peak deflection of $0.17\,\mathrm{rad}$. This improvement over Pole Placement is a direct consequence of the Bryson's rule-weighted cost function, which distributes control authority across all state variables and explicitly suppresses inter-axis coupling.

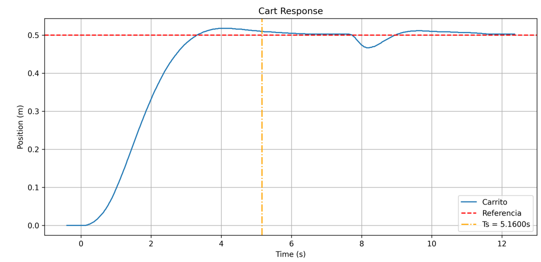

Fig. 2. Experimental step response of the cart position under Pole Placement control.

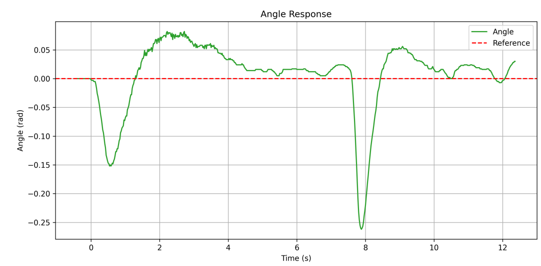

Fig. 3. Experimental pendulum deflection angle during the position step displacement under Pole Placement control.

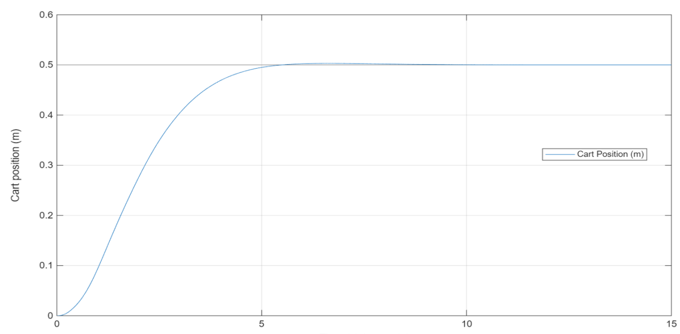

Fig. 4. Simulated cart position $p(t)$ under LQR control for a $0.5\,\mathrm{m}$ step reference.

On the physical platform, the LQR controller achieved a settling time of $t_s = 5.25\,\mathrm{s}$ and eliminated overshoot entirely ($M_p = 0\%$), as shown in Fig. 6. The pendulum angle response, shown in Fig. 7, remained within the prescribed safety margin throughout the maneuver. The complete suppression of overshoot beyond what the nominal model predicts indicates that the physical mechanical dissipation exceeds the damping captured by the linearized plant, which in this case acted beneficially by further attenuating the residual transient.

C. Comparative Analysis

Table IV consolidates the performance metrics of both controllers across simulation and experimental conditions. The settling time error remains consistent across both designs ($\Delta t_s^{\mathrm{PP}} = 14.68\%$, $\Delta t_s^{\mathrm{LQR}} = 11.68\%$), reflecting a systematic and bounded model uncertainty primarily attributable to

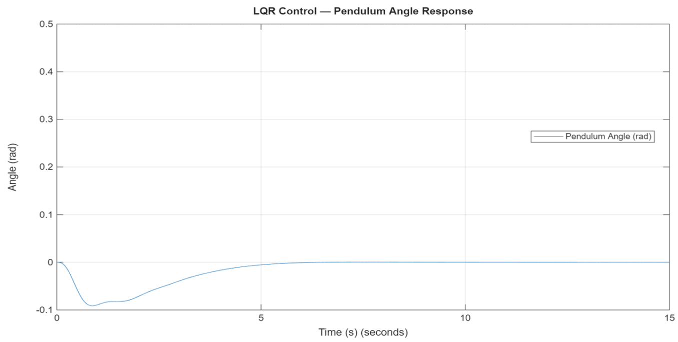

Fig. 5. Simulated pendulum angle $\theta(t)$ under LQR control for a $0.5\,\mathrm{m}$ step reference.

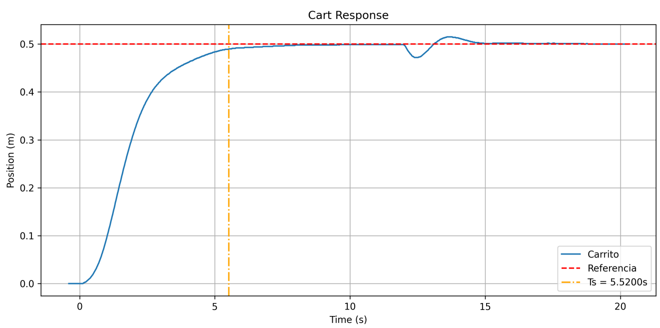

Fig. 6. Experimental cart position $p(t)$ under LQR control for a $0.5\,\mathrm{m}$ step reference.

unmodeled armature inductance and discrete-time sampling latency. The substantially larger overshoot discrepancy under Pole Placement (48.79%) confirms its higher sensitivity to unmodeled plant dynamics, whereas LQR demonstrated superior robustness by driving the experimental overshoot below the threshold of the nominal prediction.

V. Conclusions

This work demonstrated the design, simulation, and experimental validation of Pole Placement and LQR state-feedback controllers on an underactuated crane system. The empirical 4th-order SIMO model identified via the MATLAB System Identification Toolbox proved highly reliable, with a Coefficient of Variation below $2\,\%$ across independent experiments, confirming that empirical linearization provides a sound foundation for state-space synthesis [2], [5].

LQR outperformed Pole Placement across all transient metrics [1]. By penalizing state trajectories through Bryson's rule, LQR eliminated cart position overshoot entirely (M_p reduced from $3.6\,\%$ to $0\,\%$) while incurring a negligible settling time penalty of $1.7\,\%$ relative to the more aggressive PP design [3]. This performance gap extended to robustness: LQR exhibited tighter simulation-to-experiment agreement ($\Delta t_s = 11.7\,\%$) than PP ($14.7\,\%$), confirming reduced sensitivity to parametric model uncertainty. Most critically for structural integrity, LQR constrained the pendulum deflection to $\theta_{\mathrm{exp}} = 0.17\,\mathrm{rad}$, within the $\pm 0.175\,\mathrm{rad}$ safety bound, whereas PP permitted a peak of $0.20\,\mathrm{rad}$, exceeding the physical limit.

In both controllers, the Integral State Feedback architecture achieved zero steady-state tracking error ($e_{\mathrm{ss}} = 0$) in simulation and experiment, validating its effectiveness in rejecting constant input biases, mechanical dead-zones, and persistent load disturbances [4]. Collectively, these results establish LQR with integral action as the preferred strategy for precision trajectory tracking in underactuated crane systems subject to structural and safety constraints.

TABLE IV
Quantitative Controller Comparison and Model Validation

Controller	Metric	Simulation	Experimental	Error (%)
Pole Placement	t_s (s)	4.4995	5.1600	14.68
	M_p (%)	2.4194	3.6000	48.79
	e_{ss}	0.0000	0.0000	0.00
LQR	t_s (s)	4.7010	5.2500	11.68
	M_p (%)	0.6010	0.0000	—[a]
	e_{ss}	0.0000	0.0000	0.00

References

[1] K. Ogata, *Modern Control Engineering*, 5th ed. Prentice Hall, 2010.
[2] L. Ljung, *System Identification: Theory for the User*, 2nd ed. Prentice Hall, 1999.
[3] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*. Prentice Hall, 1990.
[4] G. F. Franklin, J. D. Powell, and M. L. Emami-Naeini, *Feedback Control of Dynamic Systems*, 4th ed. Prentice Hall, 1998.
[5] MathWorks, "System Identification Toolbox Documentation," Available: https://www.mathworks.com/help/ident/

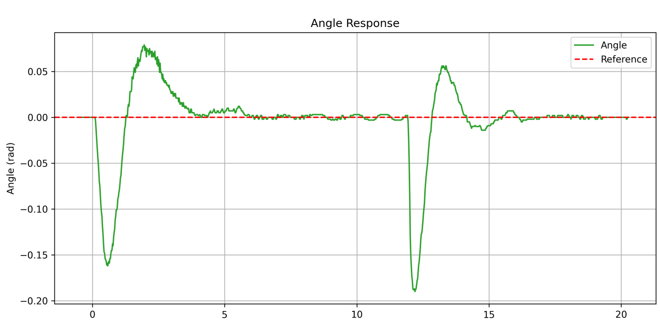

Fig. 7. Experimental pendulum angle $\theta(t)$ under LQR control for a $0.5\,\mathrm{m}$ step reference.